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MEMS 411: RECLÉM Clothes Washer Group X

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Washington University in St. Louis

JAMES MCKELVEY SCHOOL OF ENGINEERING

Mechanical Engineering Design Project

MEMS 411, Fall 2024 RECLÉM Clothes Washer

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The **RECLÉM Clothes Washer Project** addresses the challenge faced by Dr. Ruppert-Stroescu and RECLÉM in efficiently washing repurposed textiles without damaging or deforming the fabric. The goal is to design a sustainable machine that dissolves water-soluble synthetic sheets while conserving water and maintaining fabric integrity.

Performance Goals

1. Process fabric at a rate exceeding 360 in²/hr.
2. Use less than 17.5 gallons of water per cycle.
3. Limit fabric deformation to a side length change of less than 2%.

Design Evolution

Initial proof-of-concept tests using a bucket and mud mixer revealed inefficiencies in water usage and scalability. These findings led to significant design improvements, including:

- **Overhead Sprinkler System:** Reduces water waste and facilitates wastewater recycling.
- **Conveyor Belt System:** Enhances scalability, allows larger fabric patterns, and improves usability for end-users.

The revised prototype integrates a utility sink, pump, motor, and customized components to create a scalable and efficient washing solution.

Accessibility and Usability Enhancements

To ensure inclusivity, the design considers various user impairments:

- **Visual:** Tactile feedback to assess fabric cleanliness.
- **Hearing:** Visual cues to detect pump operation.
- **Physical:** Hand-operable control pedal for users with limited mobility.
- **Cognitive:** Clear warnings to prevent improper operation.

Outcomes and Future Recommendations

The RECLÉM clothes washer prototype reflects a user-centered approach while prioritizing sustainability and efficiency. Future iterations should focus on further reducing component complexity and enhancing the machine's durability and reliability in diverse user settings.

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1 Introduction

Our project aims to work with WashU's own Dr. Ruppert-Stroescu and her organization RECLÉM to promote textile repurposing and sustainable clothes design. As we learned in our interview, Reclém's original process of repurposing textiles is incredibly complicated and arduous. One step in particular, washing the re sewn clothing, has proved especially difficult. To keep the textile patterns in order, Reclém glues a water-soluble synthetic sheet to the clothing. Once the design and patterning of the item is complete, the sheet is washed off. An ongoing challenge for Dr. Ruppert-Stroescu has been to wash the clothing in a way where the plastic film is not only completely dissolved, but the clothing is also not damaged or deformed. Thus far, a reliable solution has not been found. This is the premise for our senior design project. We plan to design and build a machine that is able to effectively wash the repurposed clothing without warping the fabric.

2 Problem Understanding

2.1 Existing Devices

When coming up with a design for our device, we researched a number of existing devices for inspiration. The most relevant devices are shown below.

2.1.1 Existing Device #1: Mini Washing Machine

Figure 1 below shows two side-by-side images of an existing miniture washing device.



Figure 1: Left: Portable Washing Machine; Right: Device Dimensions and Components

Link: https://www.target.com/p/costway-5-5lbs-portable-mini-compact-washing-machine-electric-laundry-spin-washer-dryer-blue/-/A-82829555?ref=tgt_adv_xsf&AFID=google&CPNG=Appliances+-+Target+Plus&adgroup=72-4&preselect=92537155

Description: This is a mini portable washing machine. It fits with the description of our product in terms of the ability to recycle water and be more gentle with the clothes. Additionally, it has a spin-dry capability, which could be an interesting method to explore warping, if any. The clear tub will also allow us to monitor the quality of the water and know when it needs to be replaced and the effect of the biodegradable fabric on the water.

2.1.2 Existing Device #2: Washboard

Figure 2 below shows an image of a washboard made of metal and wood.



Figure 2: Galvanized Steel and Wood Washboard

Link: <https://www.homedepot.com/p/Behrens-Galvanized-Steel-and-Wood-Washboard-Large-BWBG12/317374453>

Description: This is a washboard. It fits with the description of our product because it can recycle the water into the washtub below it, and it is also extremely sustainable because it uses no electricity - human power only. The design does mean that any stretching or warping can likely be avoided, as hand-washing means one can actively change direction to counteract this. The tub below can be monitored to determine when the water needs to be replaced or filtered out.

2.1.3 Existing Device #3: Bed of Nails

Figure 3 below shows an image of the Bed of Nails acupuncture mat.



Figure 3: Bed of Nails Acupuncture Mat

Link: https://www.qvc.com/bed-of-nails-acupressure-pillow-mat-set.product.V39627.html?colorId=012&sizeId=000&ref=GBA&=&CID=GGL-SP-GGL%7CSP%7CGenericsportsfitnesspersonalcare%7CSPT%7CPLA%7CX%7CX%7CStatic%7CX%7CACQ%7CX%7CX%7CX&e4=ACQ&e5=PLA&e6=SPT&e7=17083987181&e8=CjwKCAjwufq2BhAmEiwAnZqw8h8ZawudxbVAhReP8PYnfVyvAw7G16s4cnzcFHbyozuItIMrrtPtrRoC8Z4QAvD_BwE&e9=SRCH&e10=GEN&e11=5349761714&e12=141723435043&e13=pla-1841351381906&e17=V39627-012-000&e24=g&gad_source=1

Description: This device is used for acupuncture to apply gentle pressure without actually poking or cutting the skin. Usually, it helps to increase circulation in the body. This particular model appears to have duller points, but a lot more on these little circular disks. This would be useful in our design if we used two of them to hold the clothing in place to prevent warping without actually cutting through or ruining any of the fabric.

2.2 Patents

2.2.1 Foldable washing machine (CN201762583U)

This patent describes a collapsible washing machine. It is similar to the design and construction of a typical upright washer, but the middle section has been replaced by an accordion-like design. The spinning blade can be removed, and then the top of the machine lowered all the way down as

the accordion collapses. The patent schematic is shown in Figure 4 below.

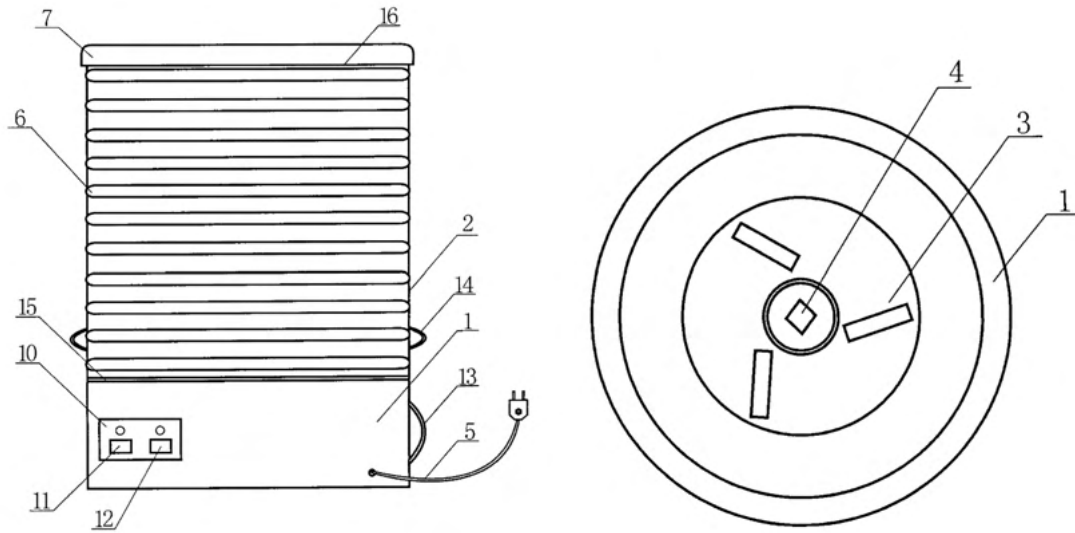


Figure 4: Patent Images Foldable Washing Machine

Each number in Figure 4 above represents a different component in the foldable washing machine.

2.2.2 Washboard (US885605A)

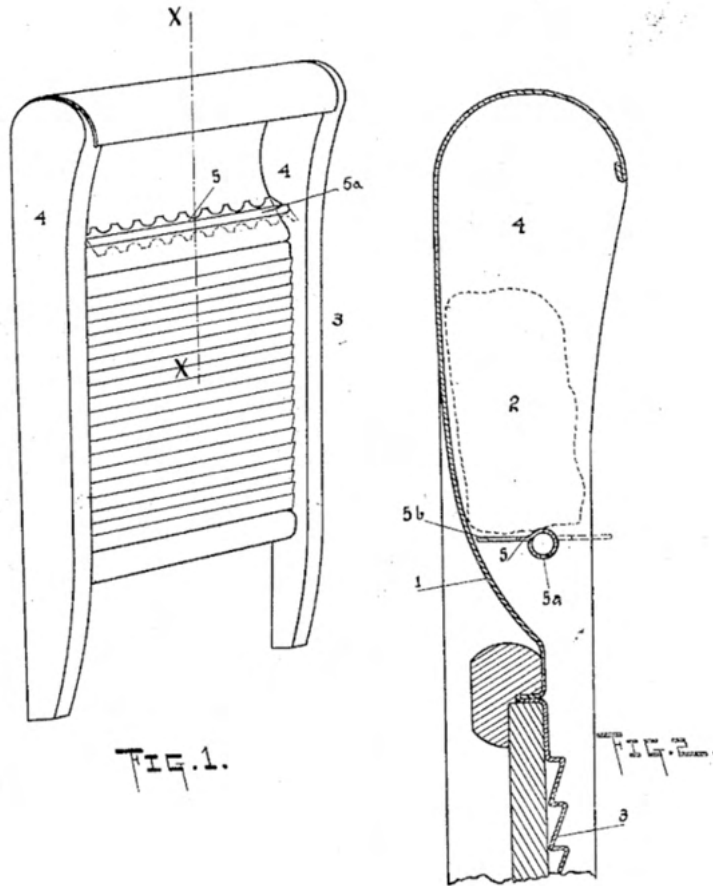
This patent details a washboard with a soap tray. It improves upon the design of a standard washboard by providing a tray that catches extra water and soap when using the board. This will provide a lot less water and soap waste than its traditional counterpart which will aid in the overall project goal of less waste. Figure 5 below shows the patent schematic of the washboard.

No. 885,605.

PATENTED APR. 21, 1908.

H. J. GILBERT.
WASHBOARD.

APPLICATION FILED MAY 12, 1905.



WITNESSES:

H. F. Tiedke
A. A. Eistler

Henry J. Gilbert INVENTOR

BY

Geo. B. Wilcox ATTORNEY

Figure 5: Patent Images Washboard

The left side of the schematic shows an outside view of the washboard device. The right side shows a more detailed side view of the inside washboard mechanism. Each component is numbered with overlapping components in both drawings numbered the same. Lastly, the witnesses, inventor, and attorney who dealt with this patent have signed the bottom of the schematic, verifying its authenticity.

2.3 Codes & Standards

2.3.1 Performance Evaluation Procedures for Household Clothes Washers (AHAM HLW-1)

This standard provides a means of comparison for the performance of non-commercial clothing washers. This standard will provide us with comprehensive definitions of each of the actions possible with a clothing washer and a quantitative way to measure the efficacy of our design. These standards will include but are not limited to: water extraction abilities, stain removal abilities (which we will adapt to how well it removes the water-soluble fabric as opposed to a stain), water usage, energy usage, safety, etc..

2.3.2 Household and similar electrical appliances – Safety – (IEC 60335-1)

This standard helps specify safety requirements for electrical appliances with less than 250 V for single-phase appliances, 480 V for appliances including direct current (DC), and battery-operated appliances meant for non-commercial purposes. The safety requirements cover everything from ways to keep dangerous components safe from the user, moisture resistance, leakage current and electric strength, markings, and classifications for the appliance, to name a few. Our device is likely to require a voltage source or battery source and we will need to guarantee its safety. This standard will be essential to understand the basic tests our device should be able to pass in order to not harm our client.

2.4 User Needs

To design a device catered to the needs of our customer, we held an interview and asked a series of questions about device requirements and various device functionalities.

2.4.1 Customer Interview

Interviewee: Dr. Mary Ruppert Stroescu

Location: Bixby Hall 14, Washington University in St. Louis, Danforth Campus

Date: September 6th, 2024

Setting: The customer introduced us to the Reclem company and went over the company's mission, motive, and method. Next a general overview of the project specifications and device constraints was given. The customer showed us samples of the clothing fabric, and gave general goals for the project. Notes were typed up on a computer. The interview was conducted in a workroom, and took ~45 min.

Interview Notes:

What is the focus of this project?

- “New, patented textile re-purposing and sustainable garment design and manufacturing”
- Reducing clothing waste
- Designing and manufacturing economically viable clothing
- Reduce production time, cost

- Enhance scalability
- Keep water preservation in mind!

What the device specifications?

- No specific requirements
- Size requirement: Eventually needs to be able to be used on a full sized coat
- Weight requirement: No specific weight requirements

What are the project goals?

- Washing clothing such that they can keep their shape
- Always achieve the same results no matter how much fabric
- Have the capacity she needs - can it accommodate a full size piece
- Take no more time than a typical load of laundry would take

General Notes

- Stretchy fabrics deform much more than other fabrics
- Adhesive spray mount sticks the clothing to the film, Store-bought adhesive hard to wash out, sticky even after washing
- Temperature of water in washing machine→ hot water (deforms the clothing)

2.4.2 Interpreted User Needs

Our customer needs table below interprets the information gathered from the customer interview and assigns an importance level for each need, where 5 is the most important, and 1 is least important.

Table 1: Interpreted Customer Needs

Need Number	Need	Importance
1	Durability	4
2	Scalability	5
3	Environmentally economical (energy/water)	5
4	Non-toxic	3
5	Production cost	2
6	Ease of use	3
7	Production time	2
8	Size of machine	1
9	Weight	1
10	Aesthetically pleasing	1
11	Ease of transport	1
12	Functionality	5

We determined the most important customer needs to be device scalability, environmental impact, as well as overall device functionality. We have also determined device durability to be fairly important as well. Our device was designed with these customer needs in mind.

2.5 Design Metrics

To create the Target Specifications table shown below, the customer needs as well as additional research into various codes and standards were done. The table presents each metric along with its associated need as well as the units of measurement and acceptable vs ideal results for that measurement.

Table 2: Target Specifications

Metric Number	Associated Needs	Metric	Units	Acceptable	Ideal
1	3	Energy efficiency ratio, as defined in 10 CFR 430.32 (g) (CFR)	lb/kWh/cycle	2	4
2	3	Water efficiency ratio, as defined in 10 CFR 430.32 (g) (CFR)	lb/gal/cycle	0.2	0.3
3	4	Textile toxicity as defined in OEKO-TEX 100	binary	pass	pass
4	5	Total cost	\$	\$400	< \$400
5	6	Usability	# of operating steps	6	> 4
6	7	Cycle time		35	< 20
7	8	Total volume	ft^2	4	< 4
8	9	Total weight	lbs	50	< 20
9	8,9,11	Ease of transport	binary	pass	pass

The design metrics shown in Table 2 above show a number of the user needs as measurable values. The Ideal column illustrates the best possible case scenario for each of our measured assisted needs. The corresponding Acceptable value illustrates a less ideal value that would still be sufficient to meet our design and customer goals. When deciding on a final design for our device, the Ideal values were taken into account.

2.6 Project Management

The Gantt chart in Figure 6 gives an overview of the project schedule.

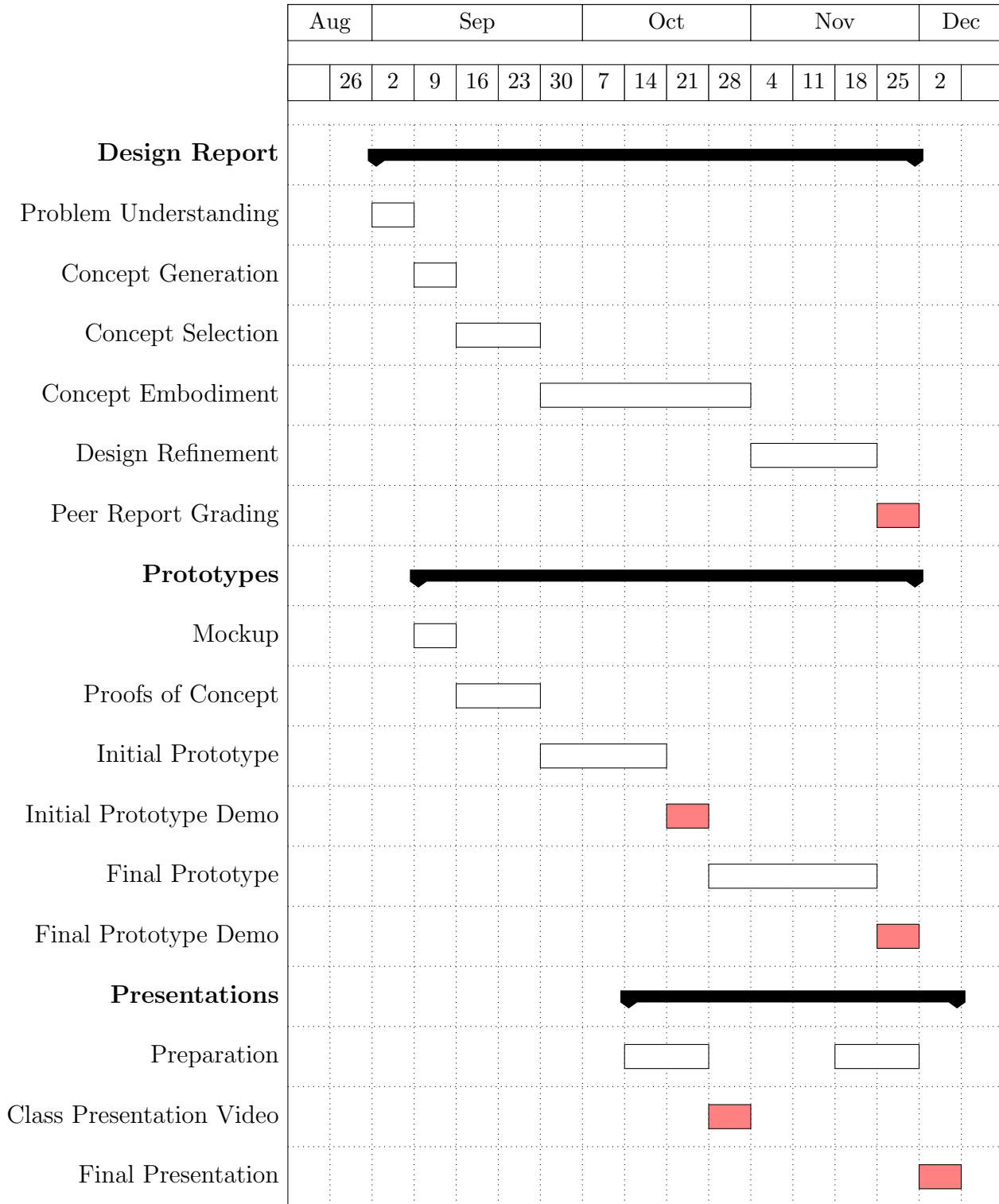


Figure 6: Gantt chart for design project

3 Concept Generation

3.1 Mockup Prototype

Our mockup is made up of primarily cardboard and hotglue. The main frame represents the housing that the clothes would be washed in. The clothes are inserted and secured into a mesh rack which is then slotted into the machine and sits on rails. The bottom of the machine is where the turbine would be located and is slanted. The slanted nature allows for the water to drain into a removable tub for disposal or reusal. Images are show in Fig.7. The mockup has made us think more carefully about how the water will be drained as well as how the material will be secured. The rack will prevent upwards pressure distortion, but the water spun by the turbine will likely cause horizontal forces that need to be accounted for to protect the clothes.



(a) Collection Tray Access



(b) Side View



(c) Top View w/ Rack



(d) Total View w/ Top and Turbine



(e) Front View

Figure 7: Mockup Images

3.2 Functional Decomposition

The function chart in Figure 8 below shows each function needed for our Reclem Clothes Washer to operate.

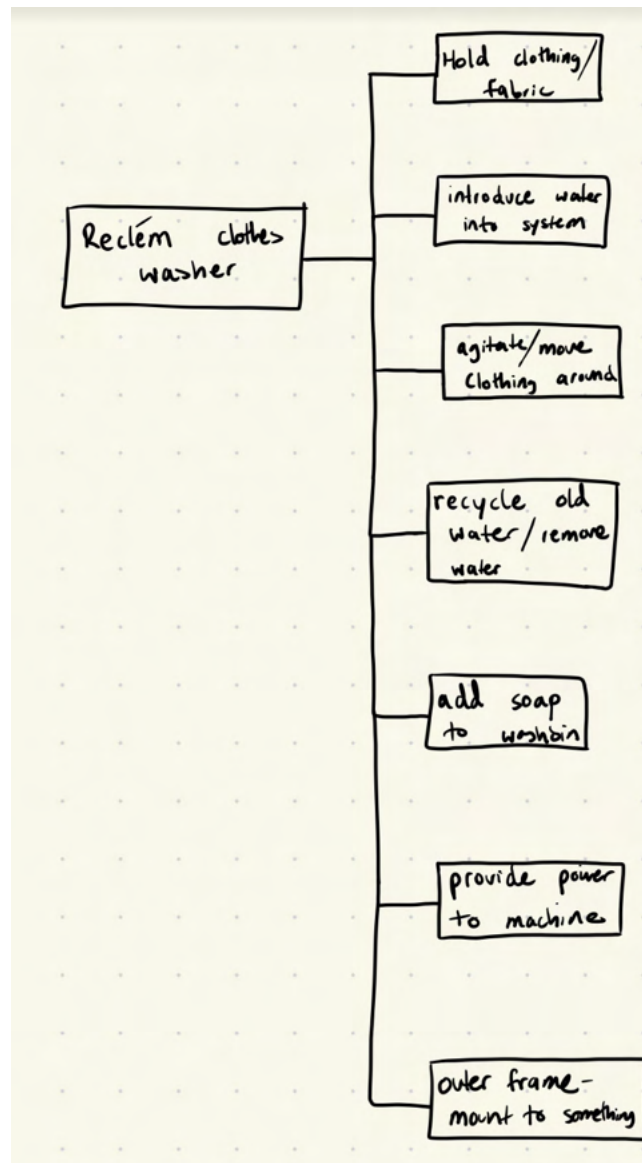


Figure 8: Function tree for Reclem Clothes Washer

3.3 Morphological Chart

The Morphological chart shown below in Figure 9 describes potential solutions for each function in the function tree.

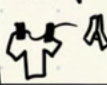
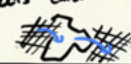
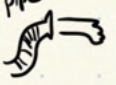
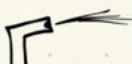
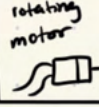
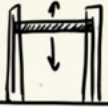
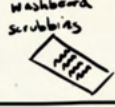
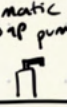
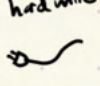
Hold fabric	pins 	clothes pins 	Pressure between 2 sheets clothes wire 	bin / bucket 
introduce water	pipe 	waterfall mechanism 	pressure sprayer 	shower head 
agitate clothing	shaker 	rotating motor 	elevator mechanism 	washboard scrubbing 
recycle water	bin at bottom to catch water, pump back to introduce water 			
add soap	automatic soap pump 	soap pipe 	add soap by hand 	
provide power	hard wire 	battery 	completely mechanical device! → no power necessary	
mount to frame	washing machine 	square basin 	cylindrical basin 	

Figure 9: Morphological Chart for Reclen Clothes Washer

3.4 Alternative Design Concepts

3.4.1 Concept #1: Turbine Agitator

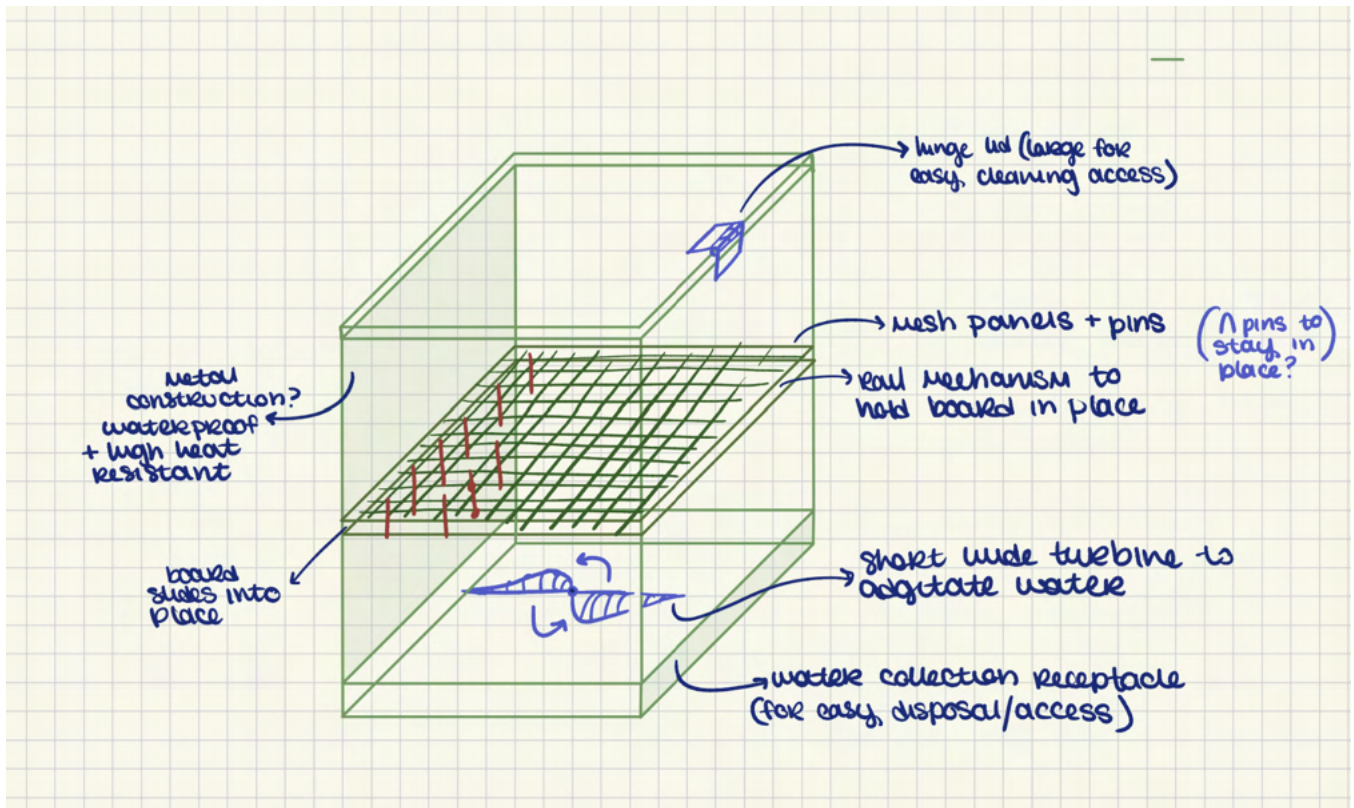


Figure 10: Sketches of Turbine Agitator concept

Description: A rectangular box with a hinged lid for easy access to the mechanism. The clothing is held in the washer using a mixture of two concepts: two plates of chicken wire to apply vertical pressure to the fabric, and pins going through the chicken wire to keep the fabric from moving laterally. The plates are stationary within the box. The box is filled with water and has a turbine connected to a motor at the bottom to agitate the water and dissolve the film covering the fabric. The bottom part of the box consists of a removable tray to remove water after the washing process is done. The water can either be disposed of or recycled, by pouring the water from the tray back into the box.

3.4.2 Concept #2: Vertical Roller concept

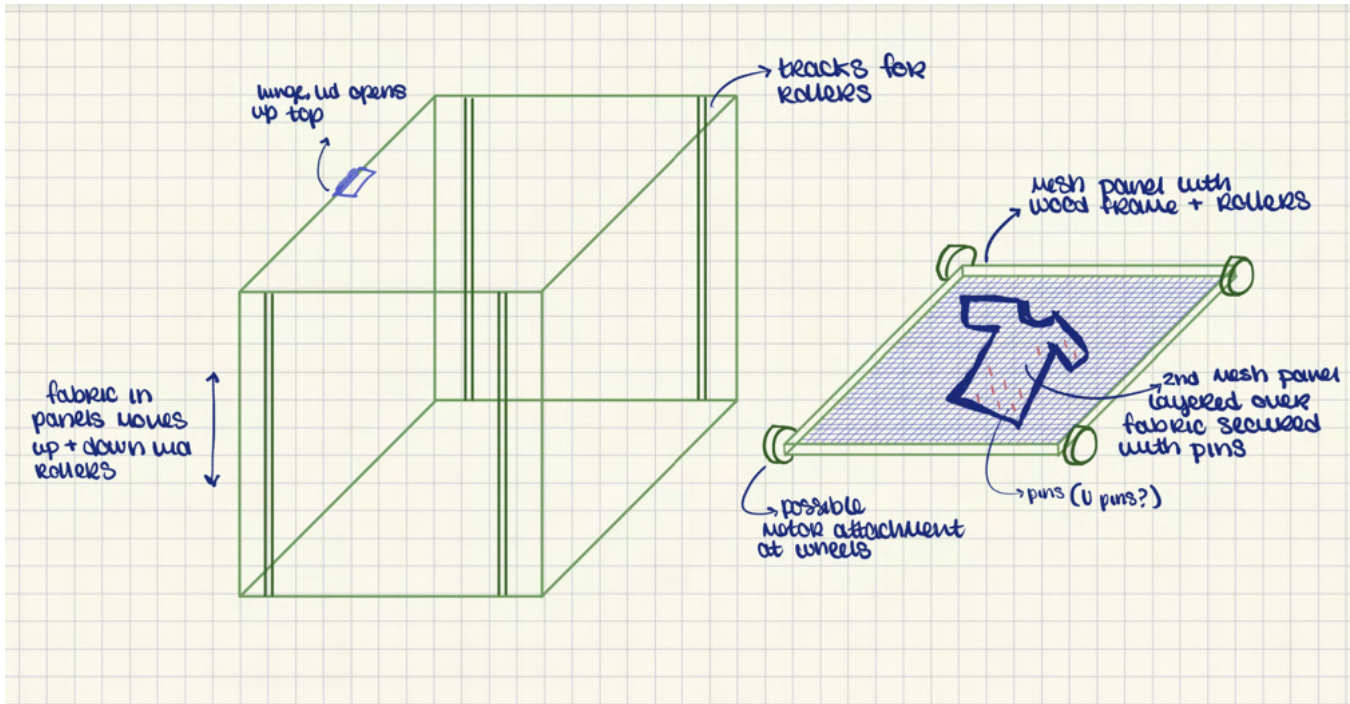


Figure 11: title

Description: This design also uses a rectangular box as the basin for the washer. There are vertical sliding rollers on two of the sides of the box. The fabric is held in place similarly to the first design through two pieces of mesh chicken wire and 'U' pins. The mesh pieces are held in a frame with rollers on each side to fit into the sliders. The mesh assembly is moved up and down on the rollers using a motor. The up/down motion agitates the fabric to dissolve the film. Similarly to the first design, there is a hinged lid over the top of the washer.

3.4.3 Concept #3: Vertical Rotation with Water Recycling Attachment concept

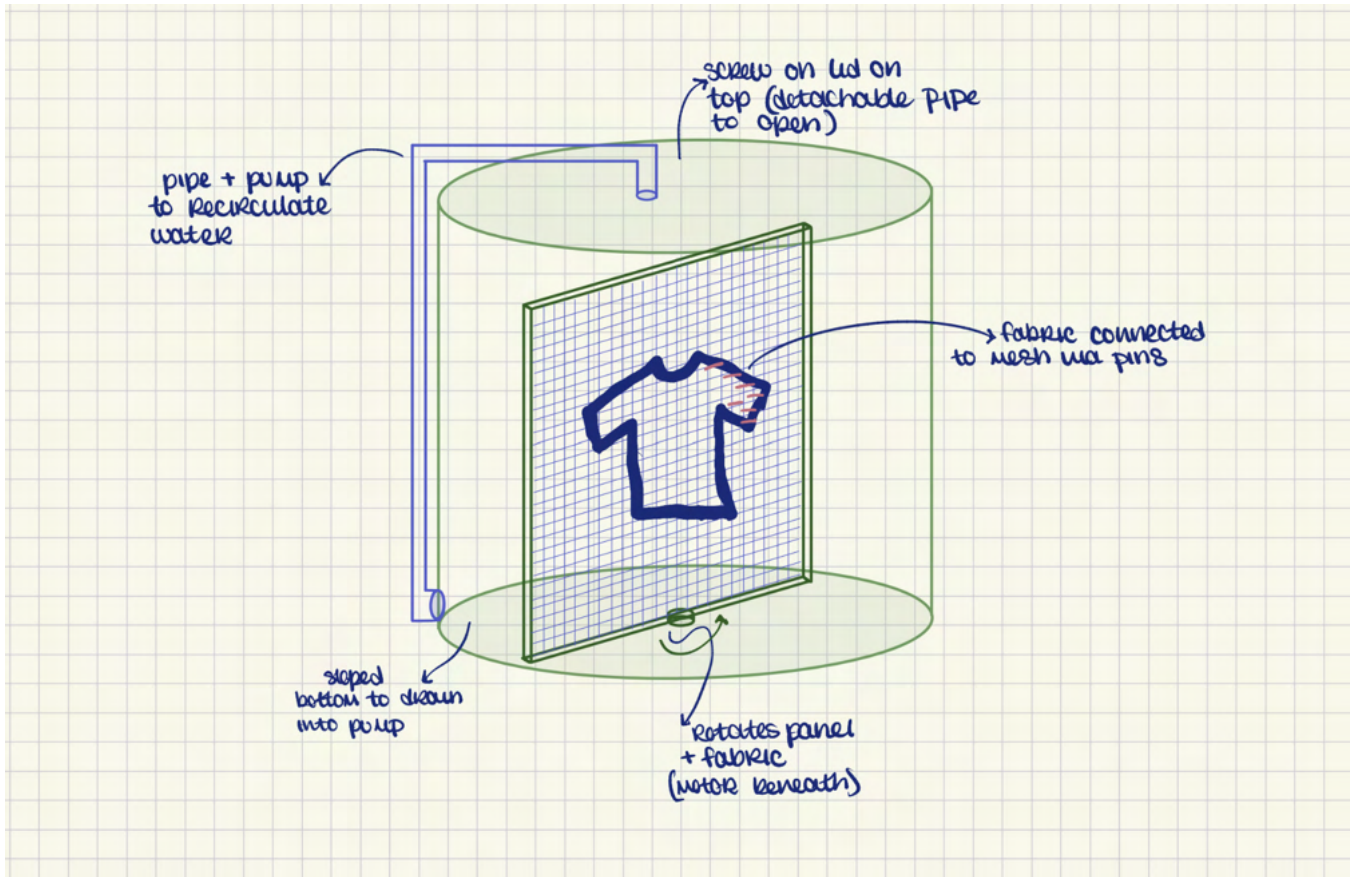


Figure 12: title

Description: The fabric is held with the same mechanisms as the previous two designs: wire mesh on either side with pins holding the fabric in place. The washer basin is cylindrical. The mesh assembly is held in the middle at the bottom of the cylindrical base. There is a motor at the junction of the mesh assembly and cylindrical base which rotates the mesh assembly. The rotation of the mesh assembly through the water filled cylindrical base agitates the water enough to dissolve the chemical film over the fabric. This device includes a water recycling system. A pump at the bottom of the basin pumps water to the top via an exterior tube. To open the basin, the pipe can be rotated out of the way for the lid to unscrew and be removed.

3.4.4 Concept #4: Lateral Rotation concept

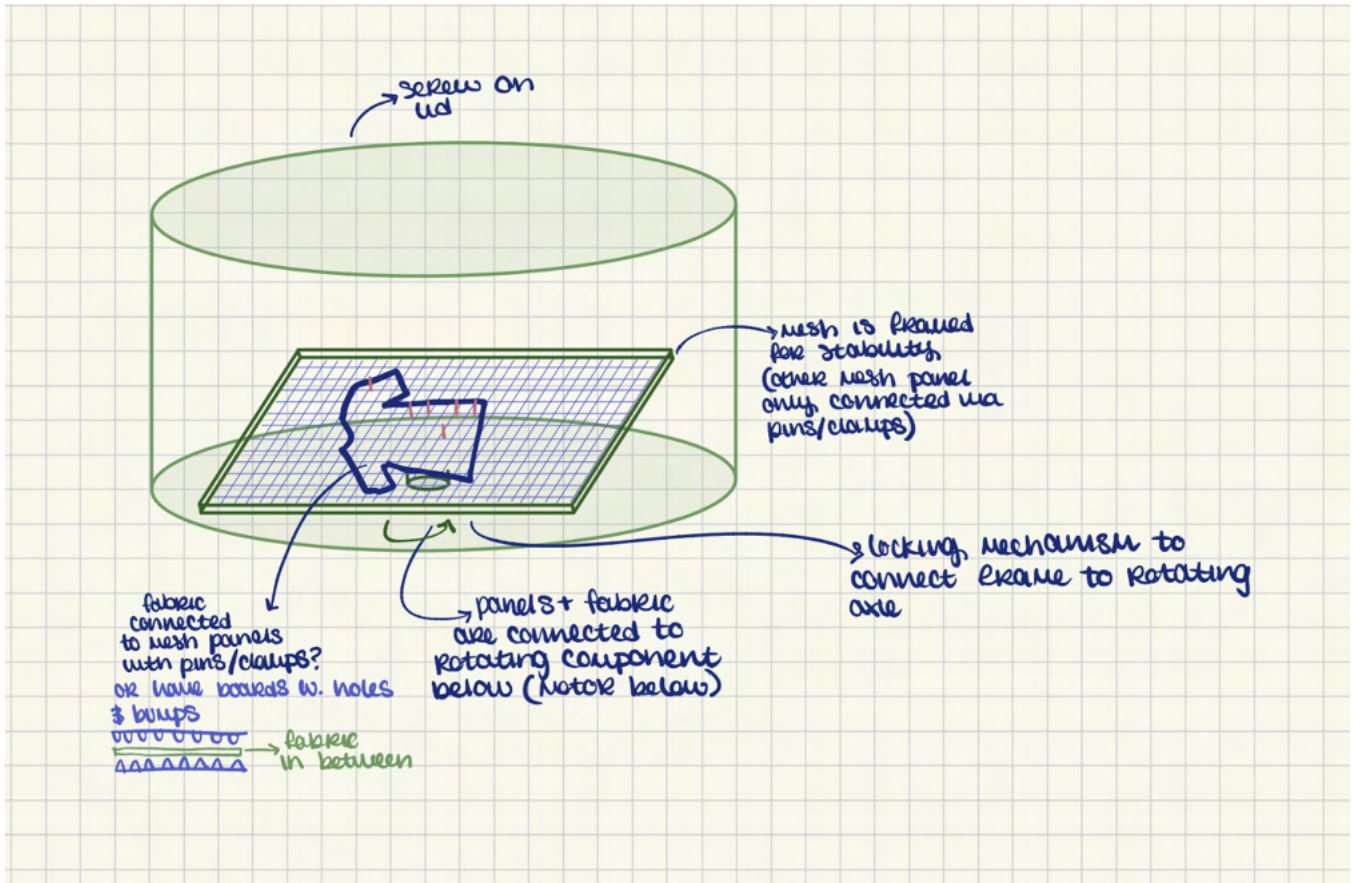


Figure 13: title

Description: The lateral rotation concept is slightly similar to the vertical rotation concept in that the mesh assembly is the moving mechanism. The fabric is held in the same way as concepts 1-3. The wire mesh and 'U' pins will hold the fabric in place as it rotates in the cylindrical basin. There is a locking mechanism to press the two pieces of chicken wire together to create pressure between the pieces. There is a motor attached to the bottom face of the mesh assembly. The lid of the assembly can be screwed off. Because the mesh assembly is only being rotated laterally, the water basin can be relatively shallow, saving water and energy.

4 Concept Selection

4.1 Selection Criteria

Below is the analytic hierarchy process (AHP) to determine the importance of design criterion. Each criterion is the same importance as itself, so it receives a 1.00. A criterion that is more important than the one it is compared to receives a whole number, while a criterion that is less important receives a fraction. These are then used to determine the percent weight the criterion should have in the design process and can be used to prioritize aspects or justify choices.

	Portable	Removes Fabric	Stops Warping	Saves Water	Fast		Row Total	Weight Value	Weight (%)
Portable	1.00	0.14	0.14	0.20	0.33		1.82	0.03	3.15
Removes Fabric	7.00	1.00	1.00	5.00	5.00		19.00	0.33	32.85
Stops Warping	7.00	1.00	1.00	7.00	7.00		23.00	0.40	39.77
Saves Water	5.00	0.20	0.14	1.00	0.33		6.68	0.12	11.54
Fast	3.00	0.20	0.14	3.00	1.00		7.34	0.13	12.70
	Column Total:						57.84	1.00	100.00

Figure 14: Analytic Hierarchy Process (AHP) to determine scoring matrix weights

4.2 Concept Evaluation

Figure 15 depicts the rankings of each of our four concepts using the new weight scale we created in fig. 14.

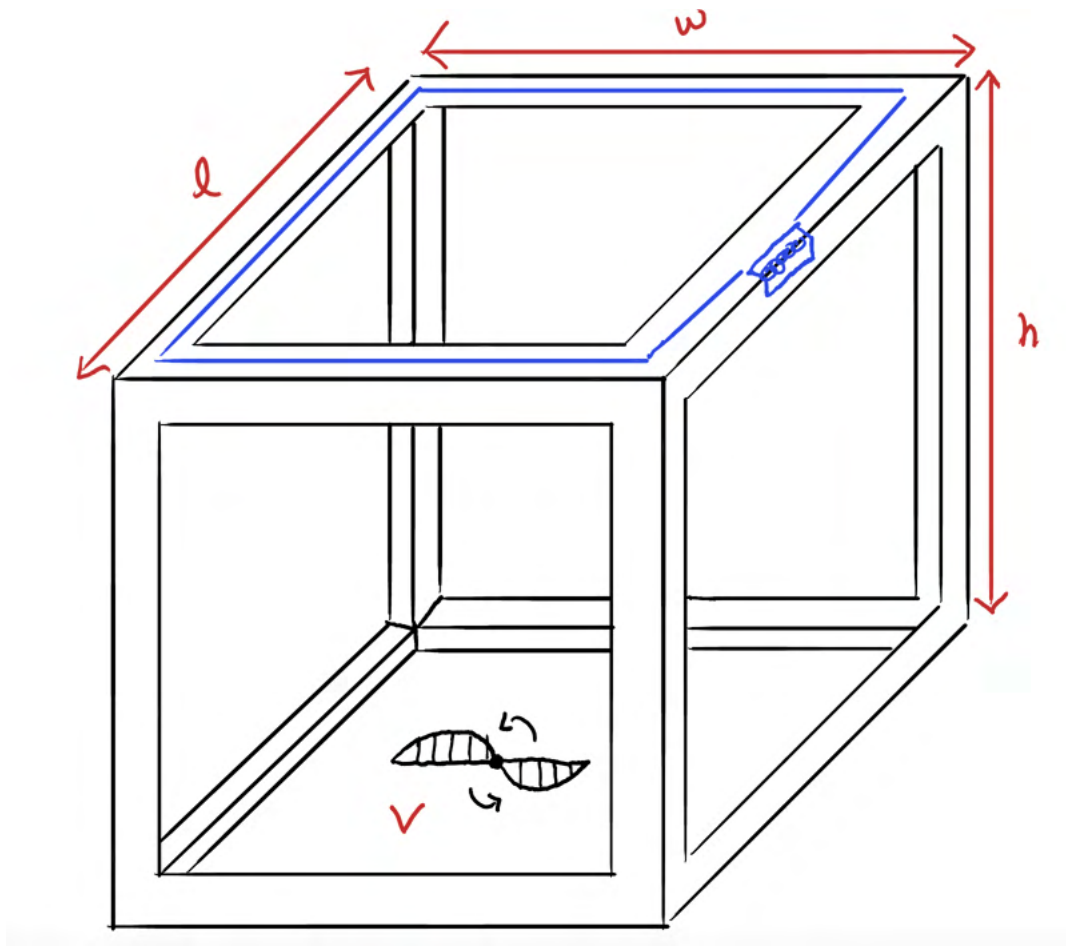


Figure 16: A model of the machine for washing textiles. The variables l , w , and h represent the dimensions of the washer, which depend on the capacity we need to meet. V represents the rotational speed of the propeller needed to agitate the water.

This model can help us make a design on what dimensions to maximize and minimize for our washer. Because an increase in textile size requires a larger washer, it is important we are careful to still be efficient about the quantity of water we need. A large textile area can translate into an even larger volume of water, since volume is cubic.

Figure 17 depicts a model of the mesh device that will hold the textile in place.

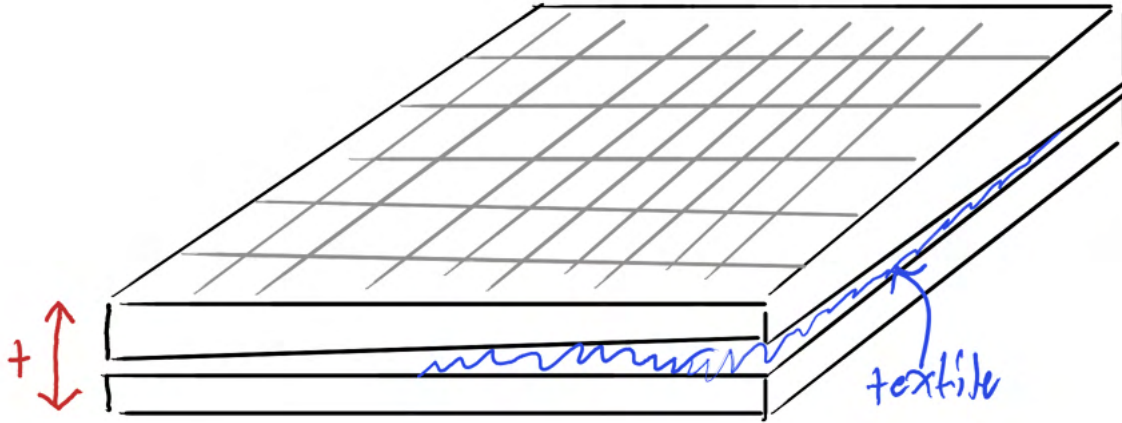


Figure 17: A model of mesh holding the textile in place. Variable t represents the spacing between the two halves of the mesh, which is dependent on the fabrics and number of layers in each piece.

This model can help us design our mesh to fit the largest number of possible fabric layers and thicknesses. We also have to be conscious of the spacing so that water can adequately flow, while also still securely holding the fabric in place.

Figure 18 depicts a model of the turbine and motor placement in the larger container and the equations necessary for the motor requirements.

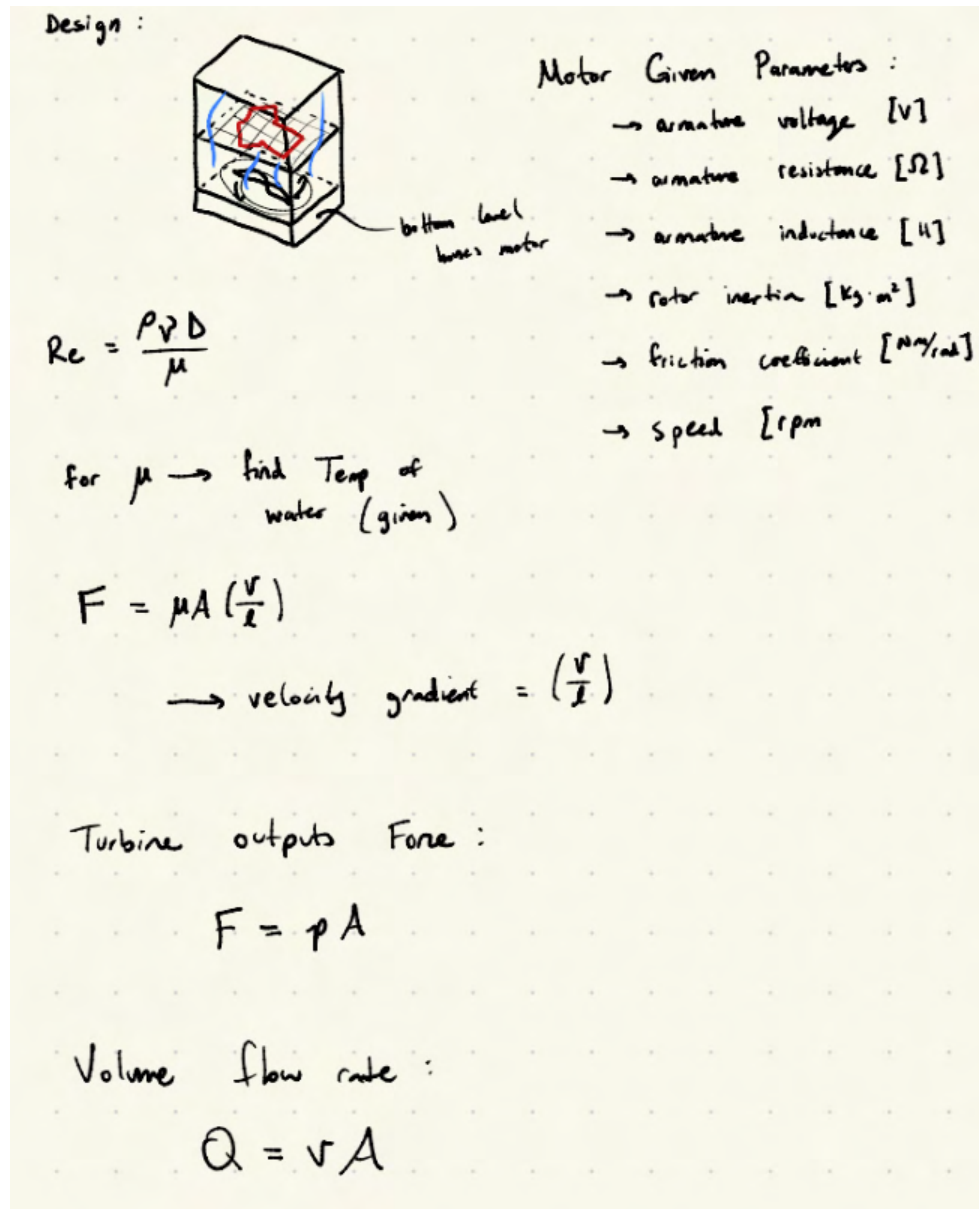


Figure 18: A model of mesh holding the textile in place. Variable t represents the spacing between the two halves of the mesh, which is dependent on the fabrics and number of layers in each piece.

The above equations will be used calculate the current required for the motor to adequately create turbulence within the water. A variety of water temperatures and turbine speeds will be tested in order to find the adequate level of agitation in the water. Additionally this will depend on how difficult the water soluble material is to dissolve.

5 Concept Embodiment

5.1 Initial Embodiment

Solidworks drawings of assemblies of our initial prototype are shown in the figures below.

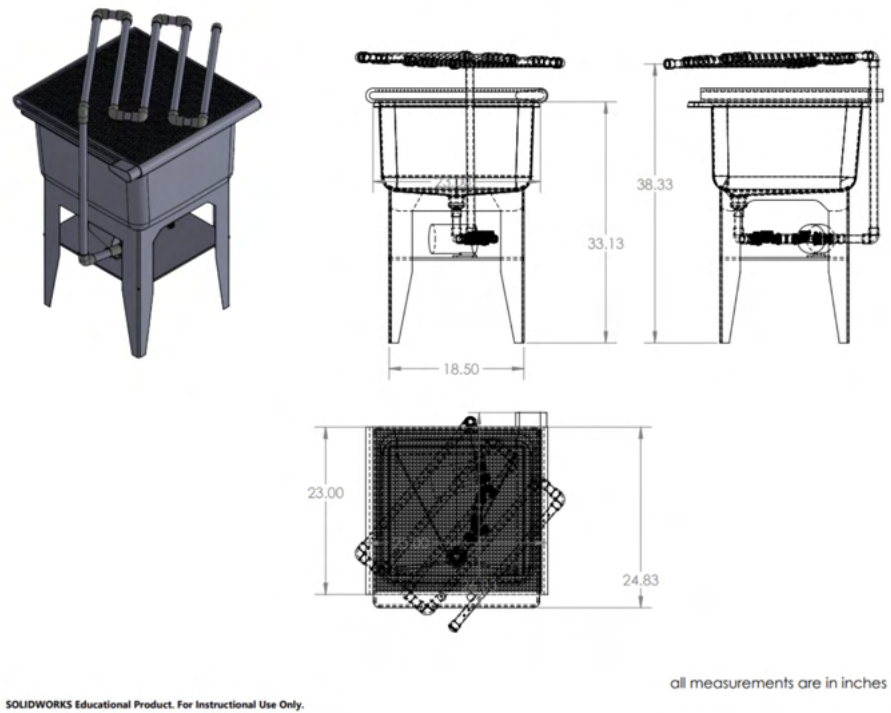


Figure 19: Assembled projected views with overall dimensions

Figure 19 shows a Solidworks drawing of our initial prototype. Front, right, and top views with general dimensions, as well as an unlabeled isometric view are included in the drawing.

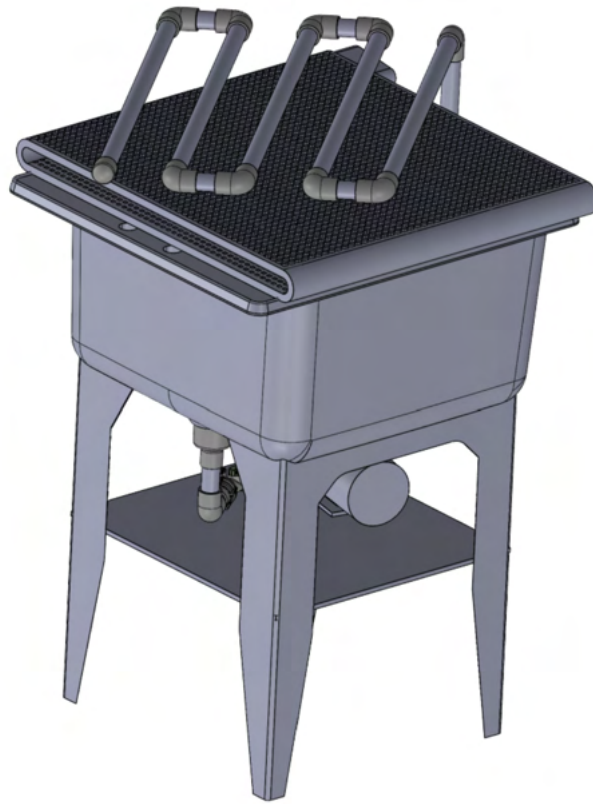
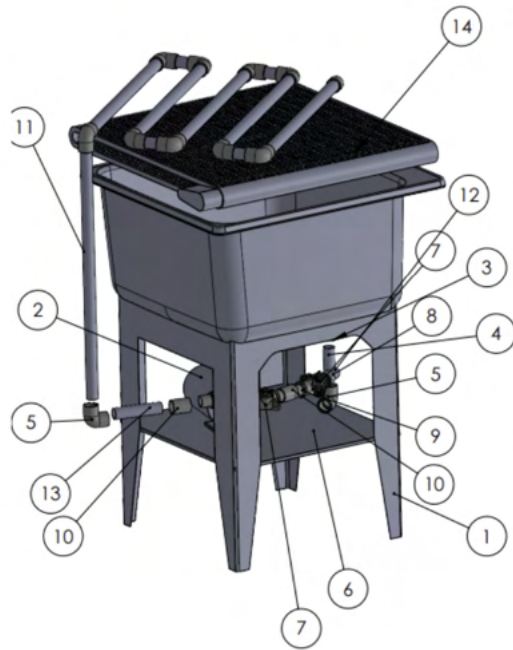


Figure 20: Large isometric view of assembly

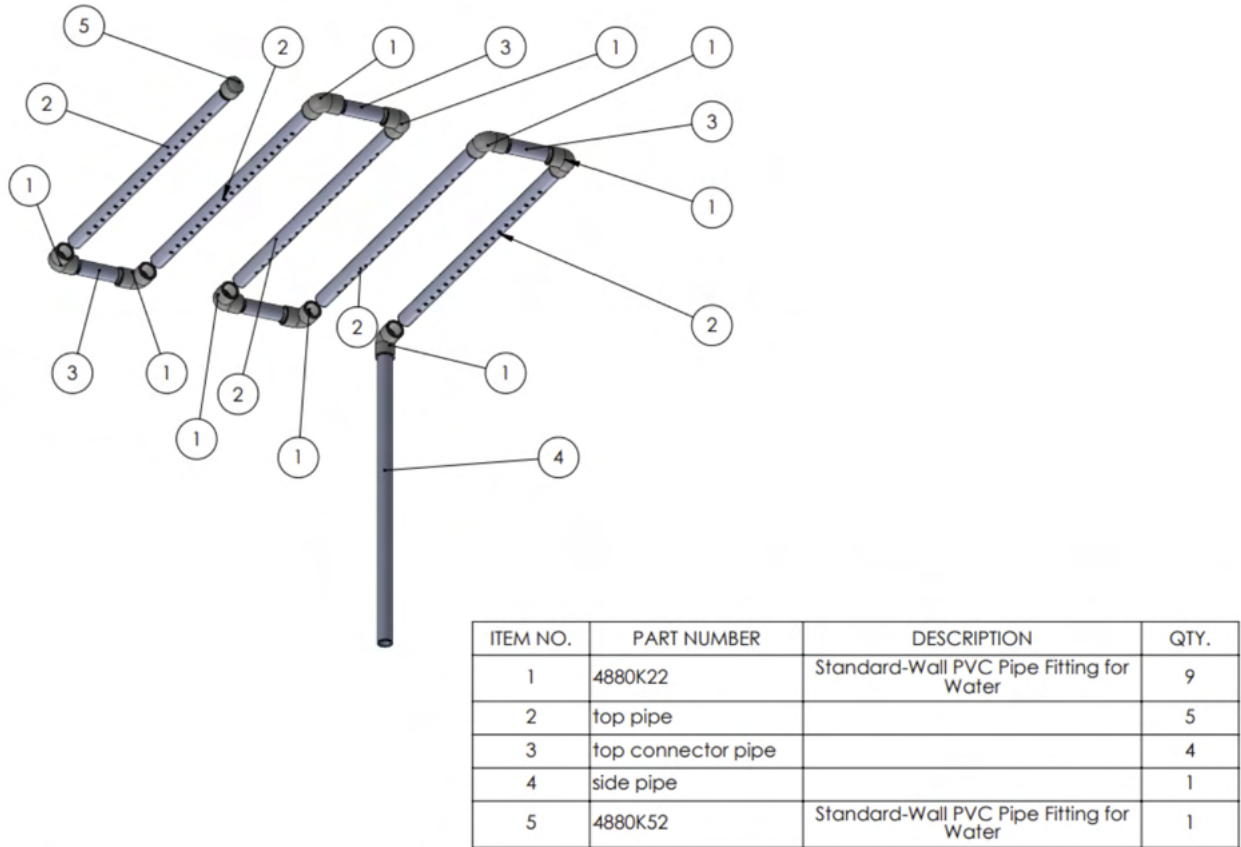
Figure 20 above shows a larger version of the completed prototype assembly from a different angle than the one included in Figure 19.



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Model14LaundryTubwithLegs-3DView-{3D}	Utility Sink	1
2	pump		1
3	4880K005	Standard-Wall PVC Pipe Fitting for Water	1
4	3in. 0.75 pipe		1
5	4880K22	Standard-Wall PVC Pipe Fitting for Water	2
6	pump support plate		1
7	6308T13_Wye-Shaped On-Off Valve	Wye-Shaped On/Off Valve	2
8	4880K169	Standard-Wall PVC Pipe Fitting for Water	1
9	4880K263	Standard-Wall PVC Pipe Fitting for Water	3
10	4880K72	Standard-Wall PVC Pipe Fitting for Water	2
11	shower head apparatus		1
12	1.5in pipe		1
13	2in pipe		1
14	conveyor		1

Figure 21: Exploded view with callout to BOM of Sink Base

Figure 21 above shows an exploded view of the complete prototype assembly with a Bill of Materials (BOM) featured in on the right hand side. Additionally, balloon callouts showing each item's BOM # are included in the drawing.



SOLIDWORKS Educational Product. For Instructional Use Only.

Figure 22: Exploded view with callout to BOM of Sprayer Apparatus

Figure 22 shows an exploded view of the water sprayer apparatus. There is also a BOM and balloon callouts for each item in the drawing.

5.2 Prototype Performance Goals & Proofs-of-Concept

5.2.1 Prototype Performance Goals

1. The device processes fabric with a throughput of $> 360 \text{ in}^2/\text{hr}$.
2. The device processes fabric using < 17.5 gallons of water per cycle.
3. After processing, rectangular pieces of fabric show a change in side length of $< 2\%$ of their original length.

5.2.2 Proofs-of-Concept

Our initial proof of concept testing was made up of a bucket filled with water, a mesh panel with a wooden frame holding the piece of fabric, and a mud mixer meant to agitate the water. However, while undergoing testing of the proof-of-concept, the team realized that the prototype was impractical. Firstly, it was discovered that a significant amount of water was required even when washing a very small piece of fabric. Additionally, the design did not look to be very scalable. After these experiments, it was thought to instead wash the fabric via a sprinkler mechanism in order to both conserve water and increase the area of fabric that the device could manage.

5.3 Design Changes

The results of our selected concept from Section 4 were very similar to the proof-of-concept design we created. None of the designs from our concept selection resembled the current design we are using for our initial prototype. The design was significantly altered in order to minimize water waste and maximize scalability in order to allow larger pattern pieces to be washed. Additionally, the initial prototype was created to improve the usability of the machine for the client. By using an overhead sprinkler connected to a drain, the overall amount of water needed is decreased and it is easier to discard the waste water compared to the selected concept. Additionally, we are able to control how much waste water to recycle through the system in order to conserve water usage. The conveyor belt controlled by a foot pedal will allow for larger pattern pieces, the ability for the user to better see the state of the fabric while it is being washed, and for the user to easily grab the fabric once the process is complete.

6 Design Refinement

6.1 Model-Based Design Decisions

Calculating the torque is a necessary step in order to ensure that the velocity of the conveyor belt is moving at an adequate speed. After calculating the torque of the motor, we will use the value to figure out how to gear up our motor in order to alter the speed of the conveyor belt. It is assumed that the current and voltage are constant and that the motor is operating at peak efficiency. Additionally, it is assumed that the pedal is pressed all the way down, outputting the maximum current. The angular momentum of the system is measured from the physical washing machine by measuring the speed of the conveyor belt. Figure 23 below describes the equations required to calculate the torque of the motor.

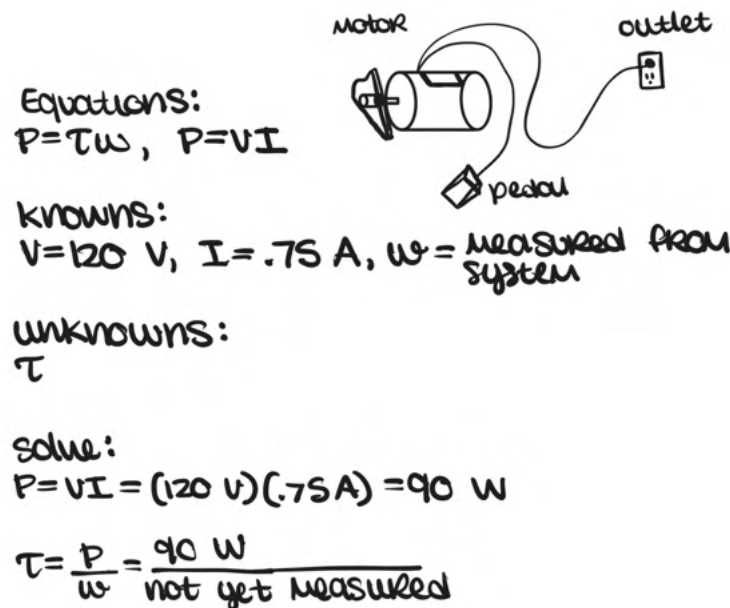


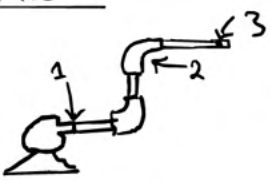
Figure 23: Model of motor torque

As our system uses a variable angular speed based on the user, we were unable to accurately calculate the torque of the motor using our model. However, we did find the power necessary to run our system to be 90W.

Figure 24 below describes the model used to calculate the velocity of the water stream coming out of the first hole on the sprayer. To find this value, we used the extended Bernoulli's Equation to account for friction losses, turbulent flow, and additional pressure losses. We assumed the pipe diameter was uniform throughout, and used k values for each elbow along the pipe system based on *Fundamentals of Fluid Mechanics* by Munson, Young, And Okiishi.

Model Based Design Rationale:
Fluid Flow from Pump to Sprayer

Model:



Equations: $A_1 V_1 = A_2 V_2$
 $P_1 + \rho g z_1 + \frac{1}{2} \alpha_1 \rho V_1^2 = P_2$
 $+ \rho g z_2 + \frac{1}{2} \alpha_2 V_2^2 \rho + \frac{\sum F L \rho}{2D} + \frac{\sum K_L \rho V^2}{2}$

Knowns:
 $z_1 = 0$ $P_2 = 0$ $\rho = 1000 \text{ kg/m}^3$ $d = 0.75'' = 0.01905 \text{ m}$
 $z_2 = 25'' = 0.635 \text{ m}$ $g = 9.81 \text{ m/s}^2$ $\mu = 1.619 \cdot 10^{-3}$
 $\nu = 1.519 \cdot 10^{-6}$
 $Q_1 = \frac{5.5 \text{ gallon}}{\text{min}} \cdot \frac{0.003785 \text{ m}^3}{1 \text{ gallon}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 3.154 \cdot 10^{-4} \text{ m}^3/\text{s}$
 $A_1 = \frac{\pi}{4} \cdot (0.75 \text{ in})^2 \cdot \left(\frac{0.0254 \text{ m}}{1 \text{ in}}\right)^2 = 2.85 \cdot 10^{-4} \text{ m}^2$

Unknowns:
 $V_1 = \frac{Q_1}{A_1} = 1.106 \text{ m/s}$ $Re = \frac{\rho v d}{\mu} = 13,878 \rightarrow \text{turbulent}$
 $\alpha_1 = \alpha_2 = 1.08 \text{ (turbulent)}$
 $P_1 = P_{\text{max capable}} = \rho g \cdot z_{\text{max}}$
 $z_{\text{max}} = 6 \text{ ft} = 1.83 \text{ m} \rightarrow P_1 = 17952 \text{ Pa}$
 $E = 0 \text{ (table 8.1)}$ $E/D = 0 \text{ (Fig. 8.20)}$
 $f = 0.025 \text{ (Fig. 8.20)}$ $\sum L = (25 + 3)'' = 0.7112 \text{ m}$
 $K_L [\text{regular } 90^\circ \text{ Flanged elbow}] = 0.3 \text{ (Table 8.2)}$

1 → 2

$$V_2 = \left[\frac{P_1 + \rho g z_1 + \frac{1}{2} \alpha_1 \rho V_1^2 - P_2 - \rho g z_2}{\left(\frac{1}{2} \alpha_2 \rho\right) \left(\frac{\sum F L \rho}{2D}\right) \left(\frac{\sum K_L \rho}{2}\right)} \right]^{1/2} = 0.0127 \text{ m/s}$$

2 → 3
 $A_2 V_2 = A_3 V_3$
 $A_2 = \frac{\pi}{4} \cdot (0.75 \text{ in})^2 \cdot \left(\frac{0.0254 \text{ m}}{1 \text{ in}}\right)^2 = 2.85 \cdot 10^{-4} \text{ m}^2$ $V_2 = 0.0127 \text{ m/s}$
 $A_3 = \frac{\pi}{4} \cdot (0.125 \text{ in})^2 \cdot \left(\frac{0.0254 \text{ m}}{1 \text{ in}}\right)^2 = 7.917 \cdot 10^{-6} \text{ m}^2$

$V_3 = \frac{A_2 V_2}{A_3} = 0.461 \text{ m/s}$

Figure 24: Model of fluid flow from pump to sprayer

In the design model shown above, the velocity of the fluid going out of the first hole in the

sprayer was found to be $0.461 \frac{m}{s}$. As the fluid travels through the sprayer, each hole it passes allows the pressure pushing the fluid to leak, resulting in a slightly lower velocity stream of water from each subsequent hole. However, due to the strength of the pump, the water is able to reach the end of the sprayer with an adequate amount of pressure to wash away any plastic film covering the fabric.

6.2 Design for Safety

Safety is an essential component of any design, particularly a design meant to be used casually by the user with added protection. In order to ensure our design is up to our standard and legal codes, we measured the risks of our machine beforehand.

6.2.1 Risk #1: Water Spillage

Description: Due to the nature of the machine requiring water and using a water sprayer in order to clean the fabric, it is possible to accidentally spray water outside of the the sink.

Severity: The severity of this risk is considered marginal due to lack of risk on human safety and only to personal property.

Probability: The probability of this risk is likely as the device always needs water to run and the sprinkler does not have perfectly machined holes, increasing the chance of spray.

Mitigating Steps: Steps to mitigate the chance of a risk include ensuring that all of the fittings are permanently connected. Additionally, ties can be added or extra methods to secure the sprinkler in place. Set up the clothing washing machine carefully in a location where it is unlikely to be disturbed/tipped over/etc.

6.2.2 Risk #2: Finger Injuries

Description: There are many small moving parts in the clothing washing machine, as a result it is likely to get a finger pinched on the convener belt.

Severity: The severity of this risk is considered to be marginal as it is unlikely that it could lead to any sever injury.

Probability: The probability of this risk is considered seldom due to the difficulty of obtaining an injury through this method.

Mitigating Steps: Mitigating steps to reduce the level of risk are to be properly trained and educated in the components of the clothing washing machine before using it. Ensuring the user is not impaired while using the machine.

6.2.3 Risk #3: Pump Overheating

Description: Due to the motorized and electrical components of the pump, it is possible for the pump to overheat if it is running continuously for a significant amount of time. Overheating the pump can lead to excess heat, smoke, and ruining the pump.

Severity: The severity of this risk is considered critical as smoke can decrease the quality of the air and there is the potential of ruining the components.

Probability: The probability of this risk is considered unlikely due to the precision with which the pump was made and the rigorous testing procedures it underwent.

Mitigating Steps: In order to minimize the risk of the pump overheating, the pump is only run after being primed and is not run continuously for more than 3-5 hours.

6.2.4 Risk #4: Electrical Contact with Water

Description: Due to the motorized and electrical components of the conveyor belt, it is possible for the water splashing off a piece of fabric to contact the motor powering the conveyor belt, as well as the wiring associated with the motor. Water contact on exposed wiring can lead to shorting of the electrical system, sparks, and possibly tripping the circuit breaker of the electrical system powering the machine.

Severity: The severity of this risk is considered critical as water damage to the motor could destroy it or impair it permanently. Electrical sparks could also be a severe danger, as they pose a shock or burn risk.

Probability: The probability of this risk is considered unlikely due to the precautions taken to shield the wiring and the motor from any splashing water. In order for water to contact the electrical components, it would either have to be spraying from an outside source, or the sprayer would have to break.

Mitigating Steps: In order to minimize the risk of the electrical contact with water, the motor and electrical connections are placed under a plastic shield, away from the main area where the sprayer functions.

6.2.5 Risk #5: Hearing Loss

Description: The pump used to move the water is not the quietest piece of machinery. Additionally, when the pump has not been correctly primed the noise levels increase even further, masking all other sounds.

Severity: The severity of this risk is considered critical as prolonged exposure to high decibel levels can cause serious damage to both the users hearing and those around the device.

Probability: The probability of this risk is considered occasional due to need to monitor the progress of the clothing during the washing cycle and the time it takes to have hearing loss.

Mitigating Steps: In order to minimize the risk of hearing damage, some sort of box or walls will be placed around the lower half of the sink to block the noise. Additionally, we recommend that once the sink process is optimized, time be spent further away from the system during the cycle.

6.2.6 Heat Map

Figure 25 highlights the risks associated with the device.

		Probability of Risk				
Severity of Risk	Category	Frequent Likely to occur immediately or in a short period of time; expected to occur frequently	Likely Quite likely to occur in time	Occasional May occur in time	Seldom Not likely to occur but possible	Unlikely Unlikely to occur
	Catastrophic					
	Critical			Hearing Loss		Electrical Contact with Water & Pump Overheating
	Marginal		Water Spillage		Finger Injuries	
	Negligible hazard presents a minimal threat to safety, health, and well-being of participants; trivial					

Figure 25: Heat map of risks.

From the figure it can be seen that the highest priority risk is hearing loss, as it is in the orange category. The next highest priority risk is water spillage. These risks will have to be mitigated properly. The other three risks: finger injuries, electrical contact with water, and pump overheating are all in the green. These should be noted, but have either already been mitigated or are unlikely enough to cause less concern. The recommendations by this heat map will be taken seriously to ensure the safety of the device.

6.3 Design for Manufacturing

We have 69 total parts in the current prototype. Currently, there are 20 threaded fasteners. Once the final pieces are installed, there will be around 30. The Theoretically Necessary Components (TNCs) are as follows: Utility Sink, Pump, Table, Conveyor Belt + Supports, Motor, Sprayer System and Tubing.

The **utility sink** acts not only as the basin to collect the used water and direct it into the drainage and recycling system, but it also acts as the housing and support for all of the other components of the prototype. Figure 26 depicts the utility sink component.



Figure 26: Utility Sink

The **pump** cannot be combined into anything else and is necessary as it operates the entire water system of the device. It is responsible for pulling the water needed to be recycled back into the system in addition to fresh water from a source. Figure 27 shows the pump setup used in the prototype.

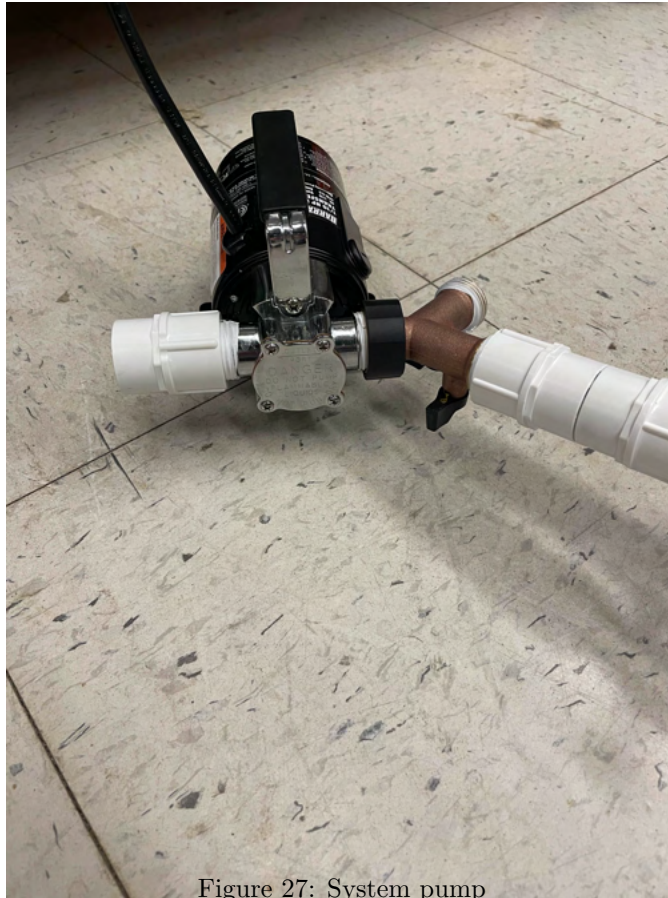


Figure 27: System pump

The **motor** is necessary to drive the conveyor belt which moves the clothes through the machine and ensures they are washed. The pump and any other components of the prototype are not capable of running a drive belt to spin gears, so this is necessary. Figure 28 depicts the motor used to drive the conveyor belt.



Figure 28: Conveyor belt driving motor.

In order to reduce the minimum number of components, the table underneath the sink could instead be a plate supported by pegs attached to the sink legs. Additionally, some of the PVC components could likely be merged into one, however, store availability affected these choices. Ideally, the changes in pipe diameter would be done with one component.

6.4 Design for Usability

Any product should be designed to consider different user impairments. Below are several considerations for the Reclém clothes washer.

6.4.1 Visual Impairment:

It could be challenging to determine the cleanliness of the fabrics when visually impaired. The washable backing material has a visually distinct “sheer” to it that is noticeably absent when washed off. To resolve this issue, users can touch the fabric samples with their fingertips to determine if it feels like the backing material is either still present, somewhat washed away, or fully removed.

6.4.2 Hearing Impairment:

The pump that powers the washing machine produces a loud noise when running. If the user has trouble hearing this noise, they could accidentally leave the pump running when it is not in use. However, this is unlikely to be an issue because when the pump is running, the water will be

spraying, which is visually apparent. Additionally, there is a distinct change in sound when the pump is running without water supplied. If left running for more than a few moments without water, the pump can start to overheat and become damaged. This can be mitigated by placing written warnings on the machine to caution the user to check if water is being supplied before the pump is powered on.

6.4.3 Physical Impairment:

We believe there would be little influence on our device from physical impairment. If someone does have limb impairment, the washer can still be used with one working arm and hand. A user with a mobility aid such as a wheelchair should still be able to use the machine, because the control pedal can be operated by hand or foot.

6.4.4 Control Impairment:

There is a potential risk of operating the washer with a control impairment. If the user is distracted or impaired, they may not realize that the water is not supplied before turning the pump on, which could lead to damage. Additionally, a user who is impaired may accidentally leave the machine running after using it, which may slowly lead to water collecting nearby from splashes.

6.5 Design Considerations

Table 3: Factors considered for design solution

Design Factor	Applicable	Not Applicable
Public Health	X	
Safety	X	
Welfare		X
Global		X
Cultural		X
Societal		X
Environmental	X	
Economic	X	

Table 4: Contexts considered for ethical judgments

Situation	Applicable	Not Applicable
Global context		X
Economic context	X	
Environmental context	X	
Societal context		X

7 Final Prototype

7.1 Overview

The final prototype consists of 4 main systems. The sprayer head, the pump system, the conveyor belt, and the motor system. These systems work in tandem to move the clothing through the machine and wash away all of the biodegradable fabric in a timely and efficient manner.

The prototype successfully met all three prototype goals of using less than 17.5 gallons of water, warping the piece less than 2 percent, and a throughput of over 360 inches squared of fabric per hour.

7.2 Documentation

Figures 29 - 33 below depict the 4 main systems and the overall final prototype.



Figure 29: The motor system. It is attached to the side of the sink by wood fixtures, and a belt connects the motor output to the conveyor belt.

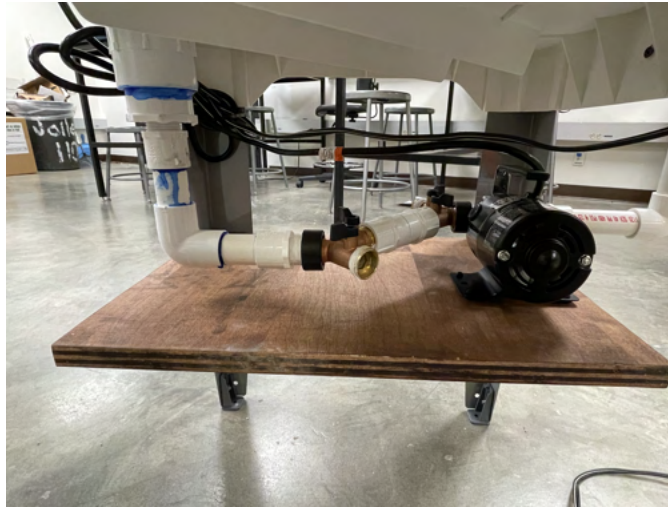


Figure 30: The piping system underneath the sink. The drainage connection is shown front and center, and the water not drained moves right towards the pump. The pump also pulls in clean water from a hose attachment on the other side. The pump then sends all the water up to the sprayer system.

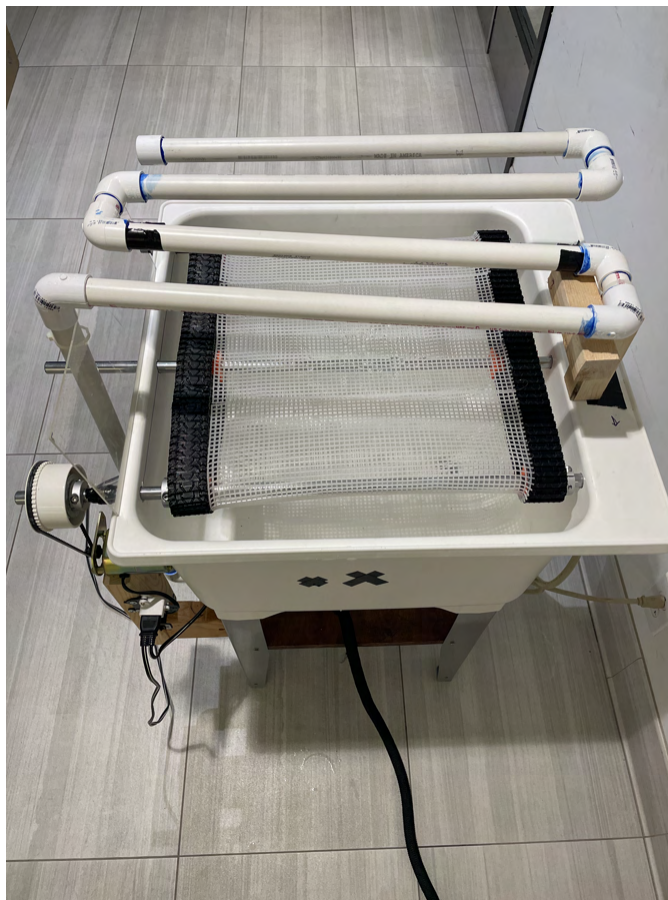


Figure 31: The sprayer system begins on the left side, with the vertical pipe seen next to the pulley. The water then flows across the conveyor belt where a piece of fabric would lay. Each piping run has sets of holes drilled along their length to spray out water.



Figure 32: The power system is shown here.

It is an extension cord that allows the user to easily turn the pump on and off with a switch rather than pulling the plug in and out from the wall socket. This also powers the motor, which has a separate switch.



Figure 33: The full washing system is shown here, including the washing sink, a drainage basin, a water supply, the sprayer, conveyor belt, and motor assemblies, and the footpedal that controls the motor.